AI HACKATHON — Prompt and Pray: Battleship Royale Official Rulebook

Field	Value
Event	AI HACKATHON
Season	Prompt and Pray
Theme	Battleship
Version	1.5
Effective	6 November 2026
Applies to	The Battleship Royale tournament: Round 1, Round 2 and the Final

> **Authority.** The Referee is the game program that runs every Battle and applies these rules the same way for every Team. If this rulebook and the Referee ever differ, the Referee's rules apply.

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1. Introduction and authority

1.1 Purpose. This rulebook sets out how Battleship Royale is played and scored and how the tournament is run. It is written for the participating Teams and for the Organizers who run and adjudicate the event. A Team new to the team page should start with Appendix C, which walks through it step by step.

1.2 Authority. The Referee is the game program that runs every Battle and applies these rules the same way for every Team. If this rulebook and the Referee ever differ, the Referee's rules apply.

1.3 Numbers. Every number a Battle uses comes from that Round's published Round Configuration. The numbers quoted in this rulebook are the ones in the event's three Round Configurations (Rule 2.9). Each Battle gives the Bot the real values, and a Bot should use the values it is given rather than numbers fixed in advance.

1.4 Terms and wording. Capitalised terms have the meanings given in Appendix A, and this rulebook uses them only in those meanings. The six results a Shot can have (Hit, Sunk, Miss, Wreck, Off the Sea and Self-hit) are defined in Rule 6.2. "Must" states a requirement, "may" a permission, and "does not" or "never" something the game does not do.

1.5 Changes effective 3 November 2026.

- (a) A Torpedo costs 7 points. It cost 6 from 2 November 2026.
- (b) A Torpedo stops at the first Ship afloat on its line, whoever owns it. It passes only the hull of the Ship that fired it and, as before, Ships already sunk. If the Ship it stops at belongs to the firer, that Ship sinks as a Self-hit (Rules 7.14 and 7.16).
- (c) The tournament takes 18 Teams: two Round 1 Groups of nine, the top six of each Group to Round 2, and the top six of Round 2 to a Final of six (Part 2). Every Round is played on a 40 × 40 Sea, and the Final's Shell speed per unit of power is now 4, as in the other Rounds.
- (d) Two new tie-breaks come before Seat letter: the earlier final point, then the earlier first point (Rule 9.5). Seat letters are drawn at random when each Battle starts (Rule 3.2).

1.6 Changes effective 5 November 2026.

- (a) The tournament takes 20 Teams: two Round 1 Groups of ten, the top six of each Group drawn at random into two Round 2 Groups of six, and the top three of each Round 2 Group to a Final of six (Part 2). This replaces Rule 1.5(c).
- (b) Every Round is played on a 36 × 36 Sea (it was 40 × 40 from 3 November 2026). The Shell speed is unchanged, so a full-power Shot still reaches about 41 Cells, more than the Sea is wide.

1.7 Changes effective 6 November 2026.

- (a) Round 2 and the Final are played on a 28 × 28 Sea; Round 1 stays 36 × 36. Six Fleets on 28 × 28 cover about as much of the Sea as ten do on 36 × 36. This replaces Rule 1.6(b) for Round 2 and the Final.

2. The tournament

2.1 Teams. Twenty Teams take part. Each Team has one Bot, a Python program that Pi, the Team's AI coding assistant, writes and changes at the Team's request.

2.2 Rounds. The tournament has three Rounds, played in this order: Round 1, Round 2 and the Final.

2.3 Round 1 Groups. Round 1 divides the twenty Teams into two Groups of ten. Teams are drawn into Groups at random.

2.4 One Battle per Group. Each Group plays one Battle. The Organizers start each Group's Battle in turn once the Round's submissions have closed, and only one Battle is live at a time. The room screen shows the live Battle.

2.5 Promotion. The top six of each Round 1 Group go through to Round 2. Those twelve Teams are drawn at random into two Round 2 Groups of six, so two Teams from the same Round 1 Group may meet again. The top three of each Round 2 Group go through to the Final, which seats those six Teams as a single Group. The Final's first, second and third places make up the podium. "Top" means the Battle's ranking order (Part 9).

2.6 The Round's team list. Each Round has a list of the Teams taking part. Only a listed Team may submit to the Round, be placed in one of its Groups, or use its workspace (Rule 11.2). Promotion adds the promoted Teams to the next Round's list. The Organizers may change the list while the Round is still a draft or is open, but may not remove a Team already in one of its Groups. A closed Round's list does not change.

2.7 Opening and closing a Round. The Organizers open each Round with a Submit Deadline, and the team page shows a countdown to it. Submissions and Armory choices close at the Submit Deadline, or earlier if the Organizers close the Round (Rules 7.4 and 11.5).

2.8 A Battle with no result. A Battle that ends as abandoned, or is reported as an infrastructure error (a failure of the event's servers rather than of any Team), shows no scores and blocks promotion, and the Organizers replay it (Rule 9.4).

2.9 Round formats.

	Round 1	Round 2	Final
Teams	All 20	12: the top six of each Round 1 Group	6: the top three of each Round 2 Group
Groups	Two Groups of 10	Two Groups of 6, drawn at random	One Group of 6
Sea	36 × 36	28 × 28	28 × 28
Seats in a Battle	10	6	6
Tick Cap	120	120	120
Shell speed per unit of power (Rule 5.1)	4	4	4
Points given to each listed Team	0	10	10
Boosters for sale	None	Flare, Cluster	Flare, Cluster, Torpedo, Teleport
Who goes on	Top six of each Group	Top three of each Group	Top three are the podium

2.10 Settings common to every Round.

Setting	Value
Fleet	carrier 5, battleship 4, cruiser 3, submarine 3, destroyer 2
Time to answer each Tick	500 ms (half a second)
Tick pacing	one Tick every 1.5 seconds, so 120 Ticks is at least three minutes of play
Silent Ticks that disconnect a Seat	3 in a row
Start-up grace	the first 5 Ticks
Physics	gravity 9.8; greatest power 5; each part of the Wind at most 1.0 either way
Flare square	7 × 7 Cells
Cluster extra Cells	up to 3
Booster prices	Flare 4, Cluster 1, Torpedo 7, Teleport 4
Scoring	1 point for each hit; a sinking bonus equal to the Ship's length; 0 points for each unhit Cell of a Seat's own Fleet

3. The Sea and the Fleets

3.1 The Sea. The Sea is a grid of Cells in rows and columns: 36 × 36 in Round 1, 28 × 28 in Round 2 and the Final. A Cell is named by its row and its column, each counted from zero. Row 0 is the top edge and column 0 is the left edge, so the Cell in row 2, column 5 is the third row down and the sixth column across.

3.2 Seats. Each Team in a Battle holds one Seat, lettered A to O. The letters are drawn at random when the Organizers start the Battle, so no Team chooses or can predict its letter. A Bot never learns which Team holds which Seat.

3.3 The Fleet. Every Seat receives the same Fleet: five Ships covering 17 Cells (carrier 5, battleship 4, cruiser 3, submarine 3, destroyer 2).

3.4 Placement. Teams never place Ships. The Referee places every Fleet on the one shared Sea, using the Battle's Seed.

3.5 Placement rules.

- (a) Every Ship is a straight horizontal or vertical line at a position chosen at random, with every free position equally likely.
- (b) No Ship overlaps or touches another Ship, diagonals included, whether the two Ships belong to one Seat or to two. The Cells diagonal to any Ship Cell are therefore always water.
- (c) Every Ship is placed independently over the whole Sea, so a Seat's Ships are not clustered and enemy Ships lie among its own.
- (d) Ships are placed longest first across all Seats: every carrier, every battleship, every cruiser, every submarine, then every destroyer. Within one Ship type, the Seats take turns in an order shuffled using the Seed, so no Seat always places first.
- (e) If the random choices ever leave a Ship with no legal position, placement starts again, continuing with the next random choices from the Seed.

3.6 Firing Origin. A Ship's Cells run from top to bottom (a vertical Ship) or from left to right (a horizontal Ship). Its first Cell, at its top or left end, is its Firing Origin for the whole Battle, even after that Cell is holed. A Teleport gives a Ship a new Firing Origin (Rule 7.21).

4. The Tick and the Action

4.1 The Tick. A Battle is played in Ticks. In each Tick, every Playing Seat receives its own private View and has 500 ms to answer with one Action. The event's servers measure the deadline; the Referee never does. Play begins when the Join Window closes (Rule 10.2).

4.2 Everyone acts at once. All of a Tick's Actions are checked, flown and resolved against one Sea: the Sea as it stood when the Tick began, with that Tick's Teleports already moved. The order of the Seats never changes an outcome; the one exception is that two Teleports in the same Tick are applied in Seat-letter order. A Ship sunk during a Tick still fires its Shell in that Tick.

4.3 Order within a Tick. The Referee resolves a Tick in this order:

1. Check every Action.
2. Use up Boosters.
3. Move teleported Ships.
4. Light Flares.
5. Fly every Shell and run every Torpedo.
6. Judge every landing Cell against that Sea.
7. Apply all damage and sinkings.
8. Score.
9. Mark disconnections.
10. Mark eliminations.
11. Check whether the Battle ends.

4.4 The Action. An Action has these parts:

Part	Required	What it gives
Ship	Yes	The name of one of the Seat's own Ships that was afloat when the Tick began
Direction	Yes	A compass bearing from 0 to 360 degrees; fractions are allowed (Rule 4.6)
Elevation	Yes	An angle from 0 to 90 degrees (0 is flat, 90 is straight up)
Power	Yes	A whole number from 1 to 5 (5 is the greatest power)
Booster	No	Flare, Cluster, Torpedo or Teleport, to fire that Booster

For example, an Action that names the carrier, direction 135, elevation 45 and power 4 fires a Plain Shell from the carrier, down and to the right.

4.5 Ship, direction, elevation and power, always. Every Action must name a Ship, a direction, an elevation and a power, even when it fires a Booster, and even when that Booster fires no Shell. A Booster the Seat has none of is fired as a Plain Shell with that Ship, direction, elevation and power (Rule 7.7).

4.6 Bearings. The direction is a compass bearing measured clockwise. 0 points straight up the map, towards lower row numbers; 90 points right, 180 down and 270 left. 360 is the same as 0.

4.7 Invalid Actions. An Action is invalid if any of these is true:

- (a) one of the four required parts is missing;
- (b) a part is the wrong kind of value, such as a word, or a yes-or-no value, where a number belongs;
- (c) a value is out of its range;

- (d) it names a Ship that is sunk or that the Seat does not have;
- (e) it asks for a Booster other than the four in Rule 4.4.

Nothing is adjusted to make an Action fit: a direction of 365 is not read as 5, and a power of 6 is not cut to 5. Anything else a Bot adds to its Action, such as a note of its reasoning, is ignored and does not make the Action invalid.

4.8 No substitution. A missing, late or invalid Action fires nothing. The Referee never substitutes or repairs an Action. If the Bot crashes or stops with an error on a Tick, that also fires nothing. Each such Tick is a Silent Tick (Part 10).

5. Aiming, flight and wind

5.1 Flight. A Bot never names a target Cell. A Shell leaves from the named Ship's Firing Origin and flies an arc. Where it comes down depends on the Action's direction, elevation and power, on the Round's gravity (9.8) and Shell speed per unit of power (4 in every Round), and on the Wind:

Quantity	How it is worked out
Launch speed	power × Shell speed per unit of power
Time in the air	2 × launch speed × sin(elevation) ÷ gravity
Landing row	Firing Origin's row + (− launch speed × cos(elevation) × cos(direction) + the Wind's row push) × time in the air
Landing column	Firing Origin's column + (launch speed × cos(elevation) × sin(direction) + the Wind's column push) × time in the air

Angles are in degrees. The minus sign in the landing row is there because direction 0 points up the map while row numbers grow downwards (Rule 4.6).

5.2 The Wind. The Wind is a steady push in two parts. The row push moves a Shell down the map when it is positive and up when it is negative; the column push moves it right when positive and left when negative. Each part is drawn at random from the Battle's Seed, with every value from −1.0 to +1.0 equally likely. The Wind is the same for every Seat and stays the same for the whole Battle. Nobody is told what it is. A Bot can learn it only by comparing where its own Shells land with where the no-wind arithmetic of Rule 5.1 says they should.

5.3 Rounding. The landing point is rounded once to the landing Cell, to the nearest whole row and the nearest whole column. A value exactly halfway between two numbers goes to the even one: 2.5 becomes 2, and 3.5 becomes 4.

5.4 Zero elevation. At elevation 0 the time in the air is zero, so the Shell lands on its own Firing Origin. This follows from Rule 5.1 and is not a special case.

5.5 Reach. Without wind, the farthest a Shell can be thrown from its Firing Origin is the square of (greatest power × Shell speed per unit of power), divided by gravity. That is $(5 \times 4)^2 \div 9.8$, about 40.8 Cells, in every Round. On the 28 × 28 Sea of Round 2 and the Final no two Cells are that far apart, so without wind any Ship afloat can reach every Cell. On Round 1's 36 × 36 Sea a Ship near a corner cannot reach the far corner, and only Ships afloat can fire, so a Seat's reach shrinks as its Ships sink.

6. What a Shot finds

6.1 The Sea a Shot meets. Each landing Cell is judged against the Sea as it stood when the Tick began, after that Tick's Teleports (Rule 4.2). "Unhit at Tick start" means not holed before this Tick. A teleported Ship keeps its holes.

6.2 Results. Every Shot reads exactly one of six results. For a Shell, including each Cell a Cluster strikes, the result is decided by this table:

Landing Cell	Result	Effect
Outside the Sea	**Off the Sea**	Nothing is hit. The Seat is still told the Cell, outside the Sea, where the Shell came down. A Wasted Shot.
Open water, including water this Seat already shot	**Miss**	A Wasted Shot.
An enemy Ship Cell unhit at Tick start	**Hit**, or **Sunk** if the Ship has no unhit Cell left after this Tick's damage	Scores (Part 8).
Any Ship Cell, anyone's, holed before this Tick	**Wreck**	Nothing changes. A Wasted Shot.
One of the Seat's own Ship Cells unhit at Tick start	**Self-hit**	Holes the Cell exactly as an enemy Shell would. No points. A Wasted Shot.

A Torpedo is not judged by this table. It follows Rules 7.13 to 7.18.

6.3 Sunk Ships are named. A Sunk result names the Ship that went down.

6.4 Sinking your own Ship. A Self-hit that holes the last unhit Cell of a Ship sinks it. The firer still reads Self-hit and earns no sinking credit. An enemy whose Shell holed that Ship in the same Tick reads Sunk and takes the sinking bonus (Rule 6.5); if no enemy Shell holed it that Tick, nobody is credited with the sinking.

6.5 Shared targets. If two or more Shells land on the same unhit enemy Cell in one Tick, each reads Hit and each firer scores the hit. When a Ship goes down, every Shell that holed one of its Cells in that Tick reads Sunk and each firer scores the sinking bonus, whether they shared one Cell or holed different Cells. The same holds when a Torpedo is what sinks the Ship.

6.6 Owner and enemy on one Cell. If an enemy Shell and the owner's own Shell land on the same Cell in one Tick, the enemy reads Hit (or Sunk) and the owner reads Self-hit.

6.7 Incoming Shells. Each View lists every Shell that holed one of the Seat's own Cells (enemy Hits, sinkings and the Seat's own Self-hits), with the result as the victim sees it and without naming the shooter. There is one entry per Shell. A Shell that lands on a Cell holed before the Tick (a Wreck) is not listed, and neither is a Torpedo's stop Cell that was already holed.

6.8 Sinking news. When a Ship goes down, every earlier Hit on that Ship in a Seat's Shot Record changes to Sunk, whoever fired the last Shell. This tells the Seat the Ship is gone, not who sank it. Scores already booked do not change: the update is news, not a sinking credit.

7. Boosters and the Armory

7.1 The four Boosters.

Booster	Price	Sold in	How it travels	A Shot?	What it does
Flare	4	Round 2, Final	Flies like a Shell	No	Lights the 7 × 7 square around where it lands
Cluster	1	Round 2, Final	Flies like a Shell	Yes	Strikes the landing Cell and up to 3 neighbours
Torpedo	7	Final	Runs straight in the Action's direction; it does not fly and the Wind does not move it	Yes	Sinks the whole of the first Ship afloat on its line
Teleport	4	Final	Fires nothing	No	Moves one of the Seat's Ships to a random free place

Buying Boosters

7.2 The Armory. Boosters are bought before a Battle, in the Armory on the team page, never during one; a Bot never buys anything. Each Round Configuration sets what is for sale, the price of each Booster, and the points every Team on the Round's list receives.

7.3 Budget. A Team's budget for a Round is the Round's points plus any points the Team carried into it. The Team may buy any number of each Booster on sale, up to 99 of one kind, as long as the total price stays within the budget.

7.4 Changing and locking a purchase. The Armory saves the Team's choice. It may be changed until the Round's Submit Deadline, or until the Round closes if that comes first. From then on it is locked, and it is what the Team's Seat holds when the Round's Battle starts.

7.5 Carrying points. Points a Team has not spent when a Round finishes carry into the next Round only if the Team is promoted into it. Boosters bought and not fired neither carry over nor come back as points. A Team that the Organizers add to a Round's list themselves, rather than by promotion, carries nothing in. Round 1 gives no points and sells nothing, so nothing carries out of it.

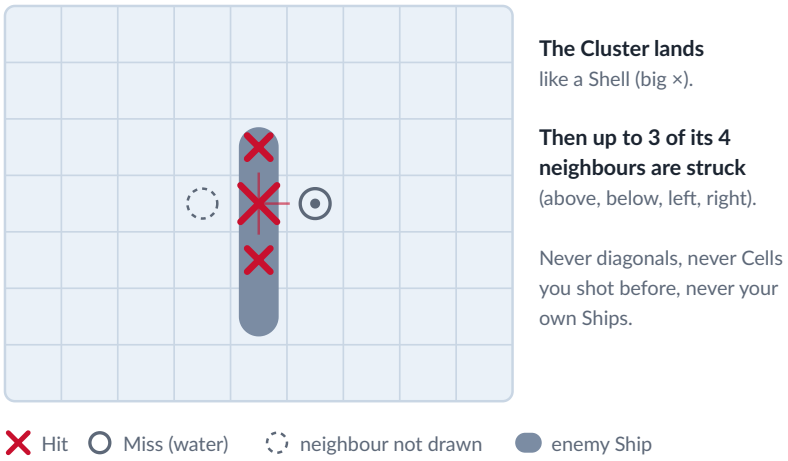
Firing Boosters

7.6 Holding and using up. Each Seat starts a Battle holding a count of each Booster its Team bought for the Round, and each View tells the Seat how many of each it has left. A Seat whose Team bought nothing, and every Seat in a Round that sells nothing, holds zero of each. Firing a Booster uses one up. Boosters left when the Battle ends are lost. The Referee refuses to start a Battle that gives a Seat a Booster the Round does not sell, or a count that is not a whole number from 0 to 99.

7.7 None left. A Booster the Seat has none of, including one not sold in the Round, is fired as a Plain Shell with the same Ship, direction, elevation and power, and the Battle's record notes that the Booster was asked for but not fired. Nothing is used up, and the Action counts as valid.

7.8 Flight is unchanged. Flares and Clusters fly exactly like a Plain Shell. They change what happens where the Shell lands, never how accurately it gets there.

Cluster

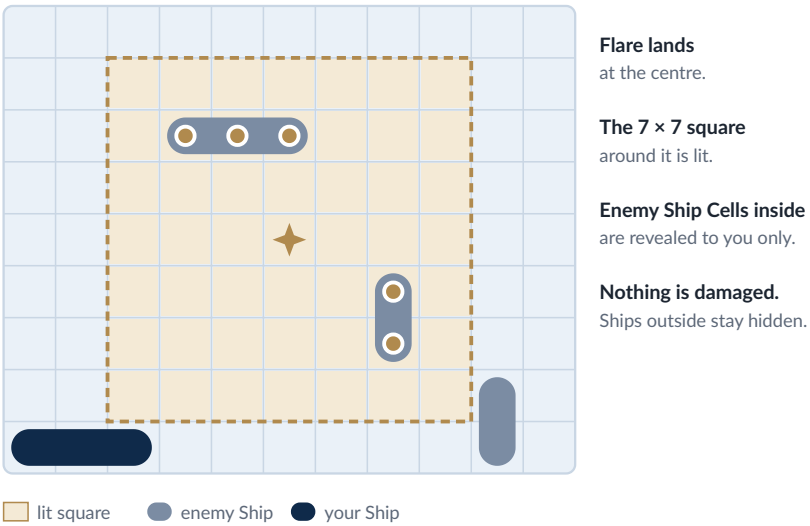


The Cluster: the main Shell lands, then up to 3 of the 4 Cells beside it are struck. A neighbour on open water is a Miss; one neighbour may not be drawn.

7.9 Cluster. The landing Cell is resolved as a Shot. Then up to 3 different further Cells are struck, drawn at random from the landing Cell's four neighbours above, below, left and right (never diagonals). Cells outside the Sea, Cells this Seat shot before this Tick, and the Seat's own Ship Cells are left out. If fewer Cells remain, fewer are struck, and one Cluster is still used up. Each extra Cell is resolved and scored like any other Shot.

7.10 Cluster off the Sea. A Cluster whose landing Cell is Off the Sea still strikes any of that Cell's neighbours above, below, left and right that lie on the Sea.

Flare

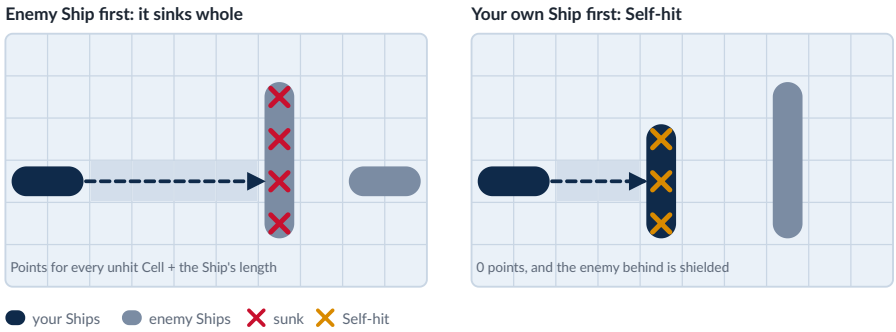


The Flare: the 7 × 7 square around its landing Cell is lit, and every unhit enemy Ship Cell inside is shown to your Team only. Nothing is damaged.

7.11 Flare. A Flare lights the 7 × 7 square centred on its landing Cell, cut off at the edges of the Sea. The Seat learns every enemy Ship Cell inside the square that was unhit at Tick start, where the Ships stand after that Tick's Teleports: a Ship that jumps into the square is lit, and one that jumps out of it is not. The lit Cells are added to the Seat's Lit Cells. Each Cell leaves the Lit Cells as soon as it is holed, or as soon as its Ship teleports away. Nobody else learns anything, not even that a Flare was fired.

7.12 A Flare is not a Shot. A Flare damages nothing, scores nothing, does not appear in the Shot Record and is never a Wasted Shot, even when it lights nothing or lands off the Sea. Its landing Cell is still given in the report on the Seat's previous Action, so a Flare also helps measure the Wind.

Torpedo



The Torpedo runs straight and sinks the first Ship afloat on its line. Left: an enemy Ship sinks whole and scores. Right: one of your own Ships is first, so it sinks as a Self-hit, scores nothing and shields the enemy behind it.

7.13 Torpedo path. A Torpedo does not fly. Its direction may be any bearing a Shell may take, fractions included. Its elevation and power must still be valid but play no part. The Torpedo runs on the surface, with no Wind, in a straight line in its direction from the named Ship's Firing Origin. The line starts at the centre of that Cell and passes through every Cell it crosses. A line that passes exactly through a corner, as at 45 degrees, goes straight on to the diagonal Cell and touches neither of the two Cells beside the corner.

7.14 What a Torpedo passes and where it stops. A Torpedo passes through the hull of the Ship that fired it and over every Ship sunk before this Tick. It stops at the first Cell of any other Ship that was afloat when the Tick began, **whoever owns it**, where that Ship stands after this Tick's Teleports. A Ship that goes down during the same Tick still stops it, and a Ship that teleports in this Tick is sunk where it lands.

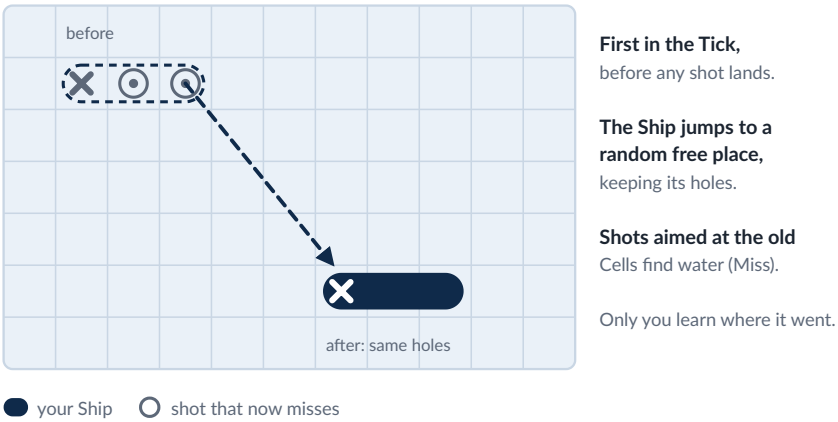
7.15 The Ship it meets sinks. Every Cell of the Ship the Torpedo stops at is holed, whether or not the Cell the Torpedo met was holed already. Against an enemy Ship, the Torpedo reads Sunk at its stop Cell, and its result also lists every other Cell of that Ship that it holed (each one not holed before this Tick), each also Sunk.

7.16 Torpedo on the firer's own Ship. If the first Ship on the line is one of the firer's own, it sinks in the same way, but every Cell the Torpedo lists reads Self-hit. It scores no points, earns no sinking credit, and counts as one Wasted Shot for each Cell listed. An enemy whose Shell holed that Ship in the same Tick still reads Sunk and takes the sinking bonus. An own Ship in the line therefore shields every Ship beyond it from that Torpedo.

7.17 Nothing in line. A Torpedo with no Ship on its line leaves the Sea and reads Off the Sea at the first Cell past the edge. That is one Wasted Shot.

7.18 A Torpedo is a Shot. In the Shot Record, a Torpedo's entries are its stop Cell, then each other Cell it holed. Against an enemy it scores by Rule 8.3. Every other Seat whose Shell holed that Ship in the same Tick also reads Sunk and scores its hit and the sinking bonus; a Cell such a Shell shared with the Torpedo scores for both.

Teleport



The Teleport moves your Ship first in the Tick, holes and all, to a random free place. Shots aimed at its old Cells find water.

7.19 Teleport. The Action's Ship is the Seat's own Ship that moves. The direction, elevation and power must be valid but play no part, and nothing is fired. The Teleport happens before any Flare is lit and before any Shell or Torpedo lands in that Tick. The Ship is lifted off the Sea and set down at a random position under the placement rules of Rule 3.5, with every allowed position equally likely: a straight line that neither overlaps nor touches any other Ship, sunk Ships included. Its old position is never one of the choices. If no position is free, the Ship stays where it is. Either way, one Teleport is used up. Several Teleports in one Tick are applied in Seat-letter order, each against the Sea the earlier ones left.

7.20 A teleported Ship is a target where it lands. Every Shell, Cluster Cell and Torpedo in the same Tick finds the Ship in its new place. A Shell that lands there hits it and can sink it, so a jump can put a Ship under another Seat's Shell. A Seat that teleports fires nothing that Tick.

7.21 Holes travel with the Ship. A teleported Ship keeps its holes position by position: if its first Cell was holed, its new first Cell is holed; if its second Cell was holed, its new second Cell is; and so on along the Ship. A Shell on a carried hole reads Wreck. The Ship's new first Cell becomes its Firing Origin from the next Tick.

7.22 Only the owner knows. Only the teleporting Seat learns where the Ship went, from its own Ships in its View and from the report on its previous Action. Other Seats are told nothing. A Shell on the Cells the Ship left, in that Tick or later, finds open water (a Miss). Cells a Flare lit there leave the Lit Cells, and earlier Shot Record entries that holed the Ship keep their old Cells. When the Ship later goes down, those entries still change to Sunk (Rule 6.8).

8. Scoring

A Seat's Score is its Hits plus its sinking bonuses:

> **Score = Hits + sinking bonuses**
>
> - **Hits:** +1 for every enemy Cell the Seat's Shots hole.
> - **Sinking bonus:** + the Ship's length for every enemy Ship the Seat's Shot sinks.

Example. A Bot holes 7 enemy Cells, and two of those Shots finish off a Cruiser (length 3) and a Destroyer (length 2). Its Score is 7 + 3 + 2 = **12**.

Rules 8.1 to 8.6 give the details.

8.1 Hits. Each enemy Cell a Seat's Shot holes scores +1.

8.2 Sinking with a Shell. A Shell that sinks an enemy Ship scores +1 for the hit plus the Ship's length:

Ship sunk by a Shell	Length	Points for the sinking Shell
Carrier	5	6
Battleship	4	5
Cruiser	3	4
Submarine	3	4
Destroyer	2	3

8.3 Sinking with a Torpedo. A Torpedo that sinks an enemy Ship scores +1 for each of the Ship's Cells that was unhit when the Tick began, plus the Ship's length once. For example, a 4-Cell Ship with one hole already in it scores 3 + 4 = 7.

8.4 Nothing else scores.

- (a) Unhit Cells of a Seat's own Fleet score nothing, because the Referee placed them, not the Bot. Every Round sets the points for each such Cell to 0. The Battle's result still shows how many unhit Cells each Seat has left.
- (b) A Self-hit scores nothing, whether from a Shell or a Torpedo.
- (c) Boosters never score by themselves. A Flare and a Teleport score nothing.

8.5 Wasted Shots. A Shot that reads Miss, Wreck, Off the Sea or Self-hit is a Wasted Shot. This includes a Cluster's extra Cells, a Torpedo that runs off the Sea, and every Cell a Torpedo lists when it sinks one of the firer's own Ships (Rule 7.16). A Self-hit holes the Cell, so that Cell no longer counts as unhit. Wasted Shots cost no points, but they break ties (Rule 9.5).

8.6 Elimination and disconnection. An eliminated Seat's Score stops. A Disconnected Seat earns nothing more from shooting, and its Ships stay on the Sea as targets.

9. The end of a Battle and ranking

9.1 Elimination. A Seat with no Ship afloat is eliminated at the end of that Tick. Several Seats can be eliminated in one Tick.

9.2 The end. A Battle ends at the end of a Tick in which at most one Seat still has a Ship afloat, or when the Tick reaches the Tick Cap. The Battle's result records the reason it ended: last standing in the first case, and the Tick Cap in the second. If both happen in the same Tick, the reason recorded is last standing.

9.3 Disconnected Seats still stand. A Disconnected Seat with Ships afloat counts as still standing, so the Battle goes on while its Ships remain targets.

9.4 Abandoned Battles. If no Seat is still connected, the Battle ends at once and the reason recorded is abandoned. That reason takes precedence over the other two. An abandoned Battle is treated as an infrastructure error: it shows no scores, it blocks promotion, and the Organizers replay it. A Battle that nobody joins ends this way too. The rule exists because every Seat dropping at once is more likely a fault in the event's servers than every Team's fault.

9.5 Ranking order. The Seats in a Battle are ranked by the steps below. Seats are compared step by step, and the first step on which they differ decides. A later step is used only while Seats are still level.

Step	Ranks higher
0. Joined	The Seat that joined the Battle at least once, even if it later disconnected, over a Seat that never joined.
1. Score	The higher Score.
2. Wasted Shots	Fewer Wasted Shots.
3. Last sinking	The earlier last sinking: the Tick in which the Seat last sank an enemy Ship.
4. Final point	The earlier final point: the Tick in which the Seat scored its last point, so reached its final Score.
5. First point	The earlier first point: the Tick in which the Seat scored its first point.
6. Seat letter	Alphabetical order. Seat letters are drawn at random (Rule 3.2).

On Steps 3, 4 and 5, a Seat that never sank anything, or never scored, ranks after every Seat that did.

In short: **score more, waste less, and get there sooner.**

> **Example.** Seats C and F both finish with 4 points and 115 Wasted Shots, and neither sank a Ship, so Steps 1 to 3 leave them level. Seat C scored its last point in Tick 96 and Seat F in Tick 110. Seat C ranks higher, by Step 4.
>
> | Seat | Score | Wasted Shots | Last sinking | Final point | Ranks |
> |---|---|---|---|---|---|
> | C | 4 | 115 | none | Tick 96 | higher |
> | F | 4 | 115 | none | Tick 110 | lower |

Step 6 means the order is always fully decided, so a ranking never changes between one look and the next.

9.6 Standings and promotion. The same order decides the Round's standings and who is promoted (Rule 2.5). After the Battle, the Team's workspace shows where the Team finished and, if its Bot fired nothing on some Ticks or was disconnected, how many Ticks and why.

10. Joining, silence and disconnection

10.1 Expected Seats. A Seat whose Team has an Accepted Submission is expected to join the Battle. A Seat without one keeps its Fleet on the Sea as targets, but its Team is not expected in the Join Window.

10.2 The Join Window. Play begins when every expected Seat has joined, or when the Join Window expires, whichever comes first. A Seat counts as joined only when its Bot actually connects to the Battle; the event's servers having started the Bot does not prove that it connected. An expected Seat that has not joined when play begins gives no Actions: after the start-up grace its silence disconnects it (Rule 10.5), its Fleet stays on the Sea as targets, and it ranks below every Seat that joined (Rule 9.5).

10.3 Silent Ticks. A missing Action, a late Action and an invalid Action each make a Silent Tick. Any valid Action resets the count, including a Booster the Seat has none of, which is fired as a Plain Shell.

10.4 Start-up grace. Silent Ticks during the first 5 Ticks of a Battle do not count, so a Bot that is slow to start is not dropped.

10.5 Disconnection. After the start-up grace, the third Silent Tick in a row disconnects the Seat at the end of that Tick. If the Bot's connection to the Battle closes, the Seat is disconnected immediately.

10.6 A Disconnected Seat. A Disconnected Seat receives no more Views and fires nothing. Its Ships stay on the Sea as targets (Rule 9.3), and if it joined at least once it still ranks above every Seat that never joined (Rule 9.5).

10.7 No return. Disconnection is final for the Battle. A Bot that reconnects is not given its Seat back.

11. Submissions and practice

11.1 Who may submit. Only a Team on the Round's list may submit to it.

11.2 The workspace. A Team's workspace is its private working area on the team page: Pi, the Team's Bot, and its practice. It is open only while a Round the Team is listed in is open and before that Round's Submit Deadline. At other times the team page shows a locked page.

11.3 Pi writes the Bot. Pi is the only one who writes or changes the Team's Bot. The team page does not show the Bot's code, and nobody can edit it there. To start again, the team page's More menu has Reset bot to starter, which puts the starter Bot back and removes the other Python files Pi added, and Clear chat, which starts Pi over with no memory of the conversation. Both wait until Pi has finished.

11.4 SUBMIT. Pressing SUBMIT enters the Bot exactly as it stands at that moment. SUBMIT is refused while Pi is still working, so that the finished Bot is the one entered. The team page warns, but does not refuse, when the Bot as it now stands crashed in its latest Practice Battle.

11.5 Submitting more than once. A Team may submit as often as it likes until the Submit Deadline or until the Organizers close the Round, whichever comes first. The Team's newest Accepted Submission is the one that plays the Round's Battle. A submission that is rejected, or whose check fails, leaves the previous Accepted Submission in place.

11.6 Submission badges. After SUBMIT, a badge on the team page shows how the submission stands. Every submission starts as Checking… and ends in one of three ways: accepted (the badge shows In followed by the Round's name), Rejected, or Check failed. Only an accepted submission is an Accepted Submission, and only an Accepted Submission takes a Seat.

Badge on the team page	Meaning
Checking…	The submission is waiting for a checker, or its check Battle is running.
In ‹Round name›	Accepted: this Bot will play the Round's Battle.
Rejected	The Bot did not pass the check. The message shown is the last thing the check reported, and says what was wrong.
Check failed	The event's servers broke during the check. It does not count against the Team, and the Team may submit again.

11.7 What the check refuses. Every submission is inspected before anything runs. The check refuses a Bot that:

- (a) uses any library other than those that come with Python, the four extra libraries the event provides (NumPy, SciPy, NetworkX and sortedcontainers), the event's own Bot library, and the Team's own files;
- (b) draws random numbers from either of the two Python sources whose numbers can never be made to repeat (the secrets library, or random bytes asked of the operating system);
- (c) has a file with the same name as one of the event's libraries or one of the libraries that come with Python;
- (d) has more than 256 KB of code in total across all the files added for the Team;
- (e) loads other code while the Bot is running, rather than through ordinary imports at its start, or runs or compiles text as code.

When a submission is refused, its badge reads Rejected and the team page says why, naming the library, file or feature at fault. The Team can ask Pi to fix the Bot and then submit again.

11.8 The check Battle. The Bot then plays a 20-Tick check Battle against built-in Bots. It holds some of every Booster the Round sells, so the parts of the Bot that fire Boosters are tested too. The Bot must not crash or run out of time on any Tick.

11.9 Practice. START PRACTICE plays a Practice Battle with every other Seat filled by one built-in opponent, chosen under Practice options (the arrow beside START PRACTICE):

Opponent	Level	How it plays
Random shooter	Easy	Fires at a random Cell in reach
Hunt & target	Medium	Searches at random; after a Hit, works outward from it until the Ship sinks
Parity hunter	Hard	Searches every other Cell, and fires a Flare when it has no lead

All three work out the Wind from where their own Shells land.

11.10 Practice and Boosters. A Practice Battle starts the Team's Seat, and every built-in opponent, with the Team's current Armory choice for the open Round. Nothing fired in practice is taken from what the Team bought.

11.11 Practice decides nothing. Only the Round's Battle decides standings and promotion. A Practice Battle's replay is visible only to its own Team.

11.12 Practice timing. In a Practice Battle, the 500 ms limit counts only the computer time the Bot itself spends on each Tick. In the Round's Battle, it counts the full time from sending the View to receiving the Action, less an allowance for network delay that is measured for that Bot's connection when it joins. A Bot that only just meets the deadline in practice may therefore miss it in the Round's Battle.

12. Fair play and what a Bot is told

12.1 The Bot plays alone. In the Round's Battle the Team's Bot plays on its own. Pi, the workspace and the Team cannot step in.

12.2 What a View holds. Each View is private to its Seat. It holds:

- the Tick number and the deadline;
- the Round's settings;
- the Seat letters in this Battle, fixed for the whole Battle and including Seats with no Bot;
- the Seat's own Score so far;
- the Seat's own Ships, with their Cells, which of those Cells are holed, and which Ships are sunk;
- the Shells that holed its own Cells (Rule 6.7);
- which of its Ships are afloat;
- how many of each Booster it has left;
- its Shot Record: every Shot it has fired and what each found (Flares are not included, Rule 7.12);
- the report on its previous Action, which is empty after a Silent Tick;
- its Lit Cells: every enemy Cell its own Flares lit that is still unhit.

12.3 What a View never holds. A Seat is never told other Seats' Ships, Shots, results, Scores or Boosters; which Seats are still standing; whether anyone fired a Flare or teleported; the Wind; the exact landing point before rounding; the Seed; or Team names.

12.4 How others can affect a View. Another Seat can change what a Seat sees in only two ways: by leaving a Wreck where this Seat shoots, and by sinking a Ship this Seat had hit, which changes those Shot Record entries to Sunk (Rule 6.8).

12.5 Scores come from the Referee. Scores come only from the Referee's report of the Battle. Anything a Bot writes out while it plays is only for finding faults and never counts towards a result.

12.6 Determinism. The same Round Configuration, Seed, Seats, starting Boosters and Actions always produce the same Battle, shot for shot. Placement and the Wind come from the Seed. The random choices for Clusters and Teleports come from separate sequences that are also drawn from the Seed, so the Sea and the Wind never change because the Bots play differently.

12.7 Checking a result. The Seed is never published, not even after a Battle is reported. The stored replay of a Round's Battle shows every Ship, the Wind, every Action and every landing, so any result can be checked from it. The Organizers re-run a disputed Battle from its Seed themselves.

Appendix A. Glossary

Term	Meaning
Accepted Submission	A submission that passed the check and shows the badge In ‹Round name›. Only an Accepted Submission takes a Seat (Rule 11.6).
Action	What a Bot sends in one Tick: a Ship, a direction, an elevation, a power and, optionally, a Booster (Rule 4.4).
Afloat	Said of a Ship that is not sunk.
Armory	The shop on the team page where a Team buys Boosters before a Battle.
Battle	One game of Battleship Royale between the Seats of one Group.
Bot	The Python program that plays the Battle for a Team. Pi writes and changes it at the Team's request.
Booster	A Flare, Cluster, Torpedo or Teleport, bought in the Armory and named in an Action to fire it.
Cell	One square of the Sea, named by its row and column, each counted from zero (Rule 3.1).
Disconnected Seat	A Seat that has been disconnected for the rest of the Battle (Rule 10.5).
Expected Seat	A Seat whose Team has an Accepted Submission, and which is therefore waited for in the Join Window (Rule 10.1).
Firing Origin	A Ship's first Cell, at its top or left end, from which its Shells and Torpedoes leave (Rule 3.6).
Final point	The Tick in which a Seat scored its last point in a Battle, so reached its final Score. A tie-break (Rule 9.5).
First point	The Tick in which a Seat scored its first point in a Battle. A tie-break (Rule 9.5).
Fleet	A Seat's five Ships: carrier 5, battleship 4, cruiser 3, submarine 3, destroyer 2.
Group	The Teams of one Round that play one Battle together.
Hit	The result of a Shot that holes an enemy Ship Cell unhit at Tick start, when the Ship is still afloat after that Tick's damage (Rule 6.2).
Holed	Said of a Ship Cell that a Shot has struck. A Ship with every Cell holed is sunk.
Join Window	The time before play during which expected Seats join (Rule 10.2).
Lit Cells	The part of a Seat's View that lists every enemy Ship Cell its own Flares have lit and that is still unhit (Rule 7.11).
Miss	The result of a Shot that lands on open water (Rule 6.2).
Off the Sea	The result of a Shot that lands outside the Sea, or of a Torpedo with no Ship on its line (Rules 6.2 and 7.17).
Organizers	The people who run the event and adjudicate it.
Pi	The AI coding assistant each Team works with. It is the only writer of the Team's Bot.
Plain Shell	A Shell fired without a Booster, or a Booster fired with none left (Rule 7.7).
Playing Seat	A Seat that is neither eliminated nor disconnected.
Practice Battle	A battle started with START PRACTICE against built-in opponents. It decides nothing.

Term	Meaning
Referee	The game program that runs every Battle: it places the Fleets, resolves every Tick, and scores and ranks the Seats.
Round	One stage of the tournament: Round 1, Round 2 or the Final.
Round Configuration	The published settings of a Round, from which every number in its Battles comes.
Score	A Seat's points in a Battle: Hits plus sinking bonuses (Part 8).
Sea	The shared grid of Cells on which every Fleet sits.
Seat	A Team's place in a Battle, lettered A to O. The letters are drawn at random when the Battle starts (Rule 3.2).
Seed	The secret number from which a Battle's random choices are drawn: the placement of the Fleets, the Wind, and the Cluster and Teleport draws. It is never published (Rules 12.6 and 12.7).
Self-hit	The result of a Shot that holes one of the firer's own Ship Cells unhit at Tick start, and of every Cell a Torpedo lists when it sinks one of the firer's own Ships (Rules 6.2 and 7.16).
Shell	What an Action with no Booster fires, and what a Flare or a Cluster fires: it flies an arc through the Wind.
Ship	One straight line of Cells in a Fleet.
Shot	Any Cell a Seat fires on that is judged and listed in its Shot Record: a Shell's landing Cell, each Cluster Cell, and each Cell a Torpedo lists. A Flare and a Teleport are not Shots.
Shot Record	The part of a Seat's View that lists every Shot the Seat has fired and what each found (Rule 12.2).
Silent Tick	A Tick in which a Seat's Action is missing, late or invalid (Rule 10.3).
Standing	Said of a Seat with at least one Ship afloat, connected or not (Rules 9.2 and 9.3).
Submit Deadline	The time after which a Round takes no more submissions and Armory choices are locked.
Sunk	The result of a Shot that holes an enemy Ship Cell unhit at Tick start when the Ship has no unhit Cell left after that Tick's damage, and of every Cell a Torpedo lists when it sinks an enemy Ship. Earlier Hits on a Ship that goes down also change to Sunk in the Shot Record (Rules 6.2, 6.8 and 7.15).
Team	The participants entered in the event under one name, who share one workspace and one Bot.
Tick	One simultaneous turn of a Battle.
Tick Cap	The last Tick of a Battle: 120 in every Round.
View	The private information a Seat receives each Tick (Rule 12.2).
Wasted Shot	A Shot that reads Miss, Wreck, Off the Sea or Self-hit.
Wind	The hidden push on every Shell, the same for every Seat for the whole Battle (Rule 5.2).
Wreck	The result of a Shot that lands on a Ship Cell, anyone's, that was holed before this Tick (Rule 6.2).

Appendix B. Quick reference

The day. 20 Teams. Round 1: two Groups of 10 → top six of each Group, drawn at random → Round 2: two Groups of 6 → top three of each Group → Final: 6 Teams → top three make the podium. One Battle per Group. Ranking decides promotion.

	Round 1	Round 2	Final
Sea	36 × 36	28 × 28	28 × 28
Seats	10	6	6
Points to spend	0	10 + carried	10 + carried
For sale	nothing	Flare 4, Cluster 1	Flare 4, Cluster 1, Torpedo 7, Teleport 4
Shell speed per unit of power	4	4	4

Every Tick. The Bot answers within **500 ms** with one Action. One Tick every 1.5 seconds, **120 Ticks** at most. An Action names:

- one of the Seat's Ships that is afloat;
- a direction from 0 to 360 degrees (0 up, 90 right, 180 down, 270 left);
- an elevation from 0 to 90 degrees;
- a power from 1 to 5;
- optionally, one Booster: Flare, Cluster, Torpedo or Teleport.

What goes wrong.

- A missing part, the wrong kind of value, or a value out of range → **invalid, fires nothing**. Nothing is repaired.
- A Booster with none left → a Plain Shell with the same aim.
- 3 Silent Ticks in a row (after the first 5 Ticks) → **disconnected** for the rest of the Battle.

Score = Hits + sinking bonuses. Hit +1 · Sunk +1 + the Ship's length · Miss, Wreck, Off the Sea, Self-hit → Wasted Shot, 0 points.

Boosters.

- **Flare:** lights 7 × 7. Not a Shot, never Wasted, secret.
- **Cluster:** the landing Cell plus up to 3 neighbours above, below, left and right.
- **Torpedo:** straight line, no Wind; sinks the **first Ship afloat on the line, your own included**. On your own Ship, every listed Cell is a Self-hit: 0 points, one Wasted Shot each. Nothing in line → Off the Sea.
- **Teleport:** happens first in the Tick; moves your Ship, holes and all, to a random free place. Fires nothing.

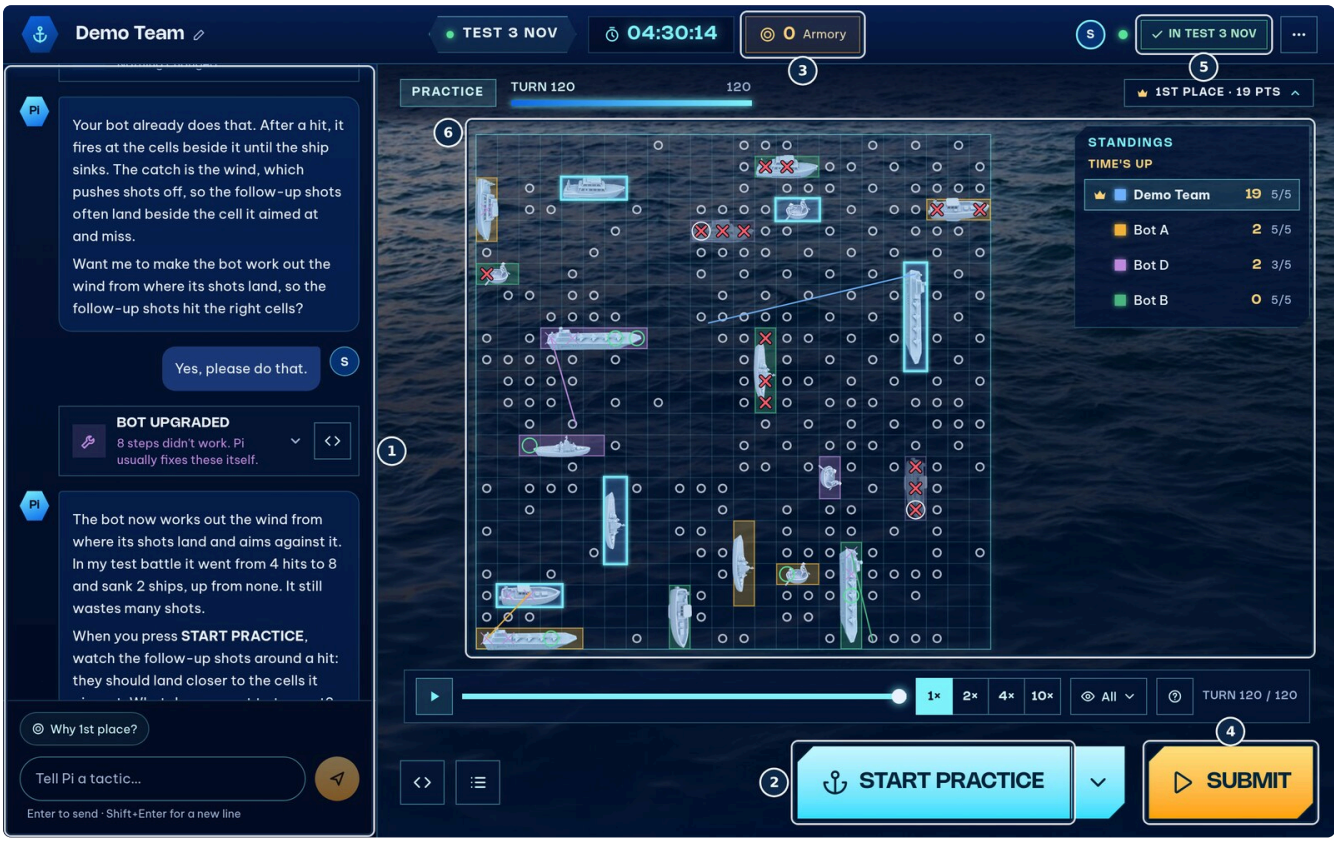
Ranking. Joined above never-joined → higher Score → fewer Wasted Shots → earlier last sinking → earlier final point → earlier first point → Seat letter (drawn at random). No sinking, or no point, ranks last on that step. *Score more, waste less, get there sooner.*

Submitting. Press SUBMIT any number of times before the deadline. The newest Accepted Submission plays. A Rejected submission or a failed check leaves the previous one in place. If a submission is Rejected, the team page says why; ask Pi to fix the Bot and submit again. Your Bot plays the Round's Battle alone.

Appendix C. The team page, step by step

The pictures show a Team called Demo Team in a test Round. The numbers in the circles are the same in every picture.

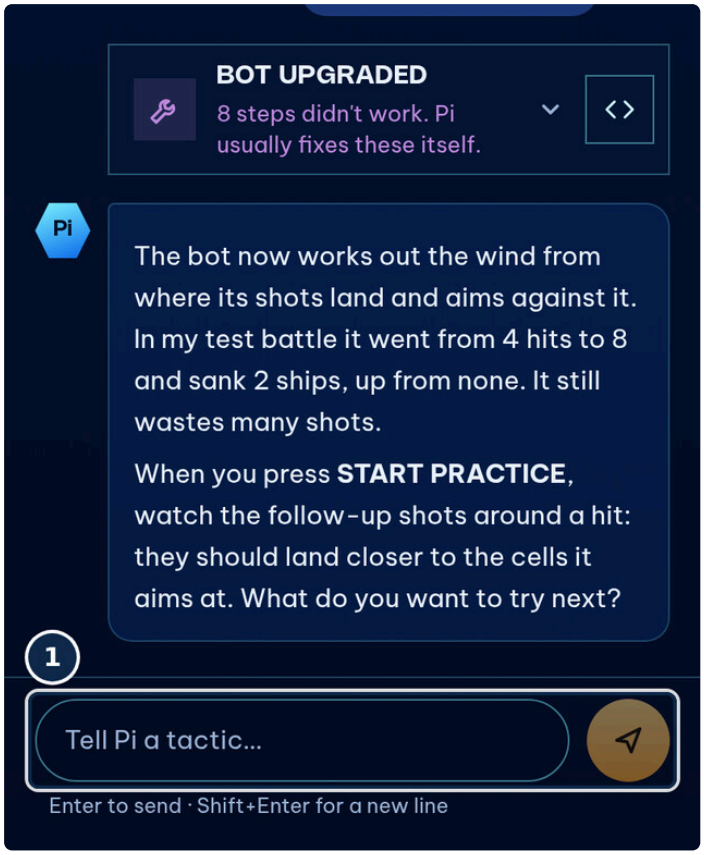
C.1 The page at a glance.



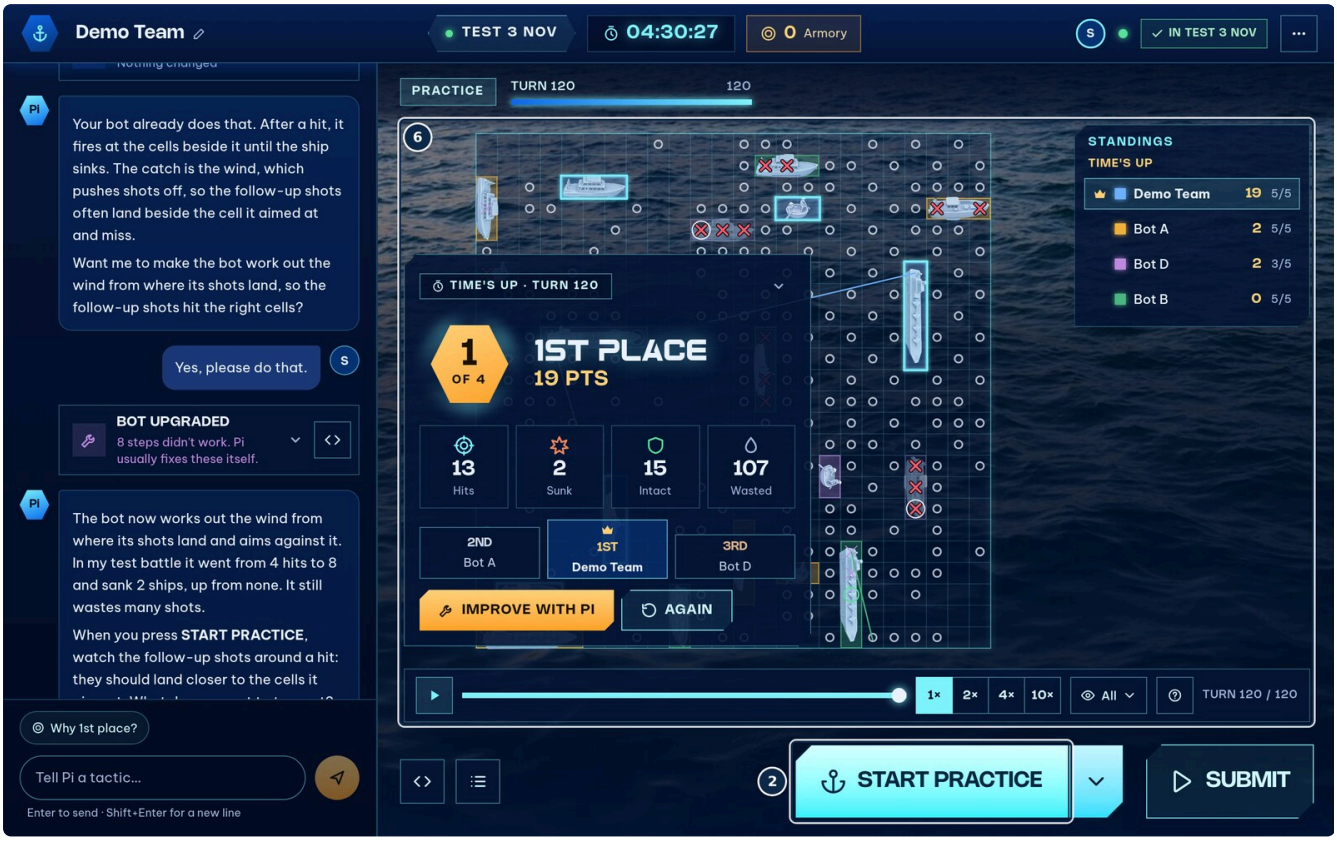
No.	What it is	What you do there
1	Chat with Pi	Describe, in plain words, how your Bot should play. Pi changes the Bot and tests it.
2	START PRACTICE	Play a Practice Battle against built-in opponents (Rule 11.9). The arrow beside it holds the Practice options: the opponents' level, and a seed to replay the same Battle.
3	Armory	Buy Boosters with the Round's points (Part 7).
4	SUBMIT	Enter the Bot as it stands now for the Round's Battle (Rule 11.4).
5	Submission badge	How your latest submission stands: Checking..., In followed by the Round's name, Rejected, or Check failed (Rule 11.6). Press a Rejected badge to see why.
6	Battle view and standings	The latest Battle: the Sea, the shots and the scores.

The clock in the top bar counts down to the Submit Deadline (Rule 2.7). When it reaches zero, the workspace locks (Rule 11.2): Pi, practice, SUBMIT and the Armory. The last Accepted Submission plays.

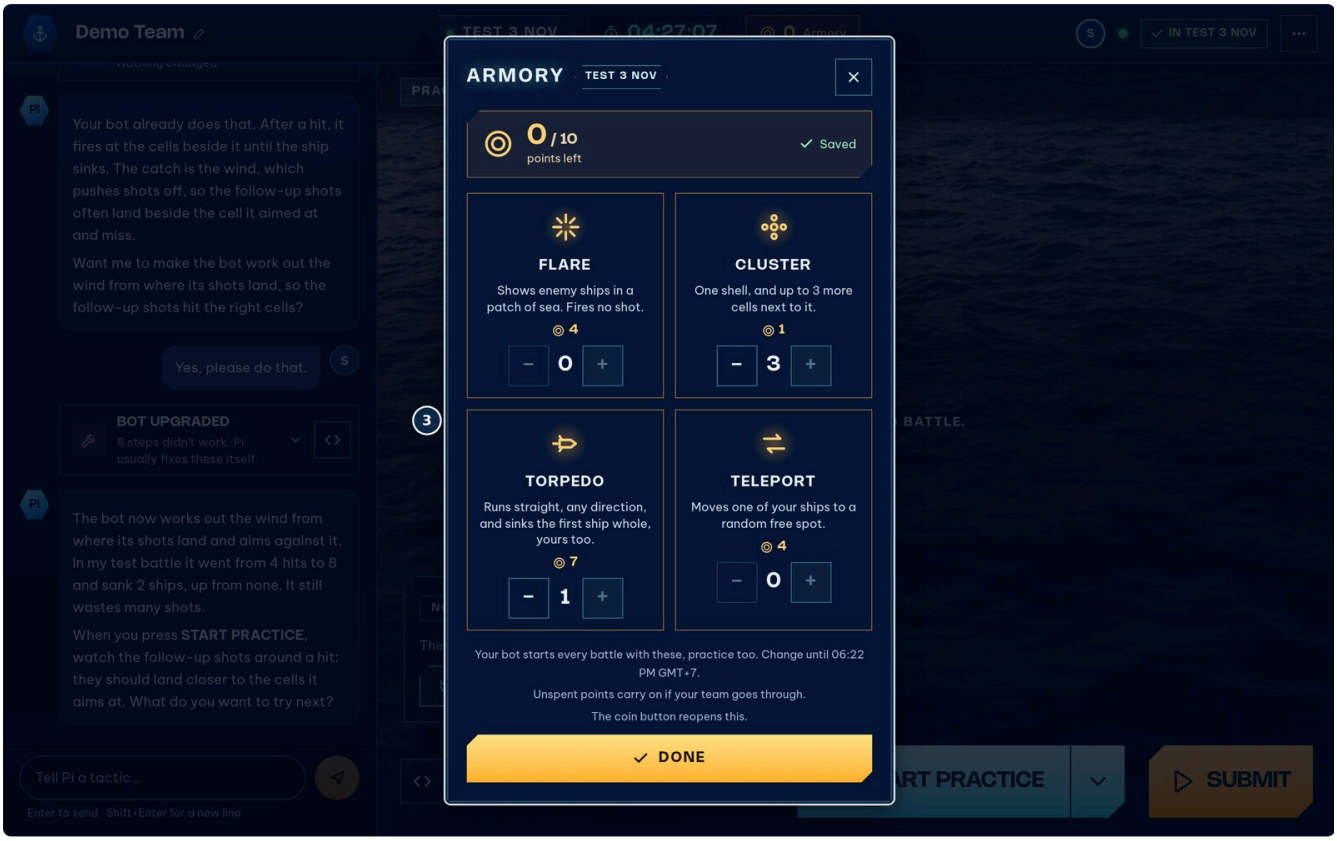
C.2 Talk to Pi. Type a tactic into the message box (1) and press Enter. Ask for one idea at a time, for example "When we hit a ship, fire at the cells next to it until it sinks." Pi explains what it will do, changes the Bot and tests it. A BOT UPGRADED card shows that the Bot changed (BOT TESTED or BOT CHECKED when nothing did); it can mention steps that did not work along the way, which Pi usually fixes itself. Only Pi changes the Bot (Rule 11.3).



C.3 Practise. Press START PRACTICE (2). The Practice Battle plays in the battle view (6), and a result card then shows your place, points, Hits, Sunk Ships, intact Cells and Wasted Shots. IMPROVE WITH PI puts a question about the result into the message box for you to send; AGAIN plays another. If the card says Your bot crashed, press Ask Pi to fix it. Practice uses your current Armory choice and costs nothing (Rule 11.10).



C.4 Buy Boosters. Open the Armory (3). Use the plus and minus buttons on each Booster; the points left are shown at the top, and the total may not go above your budget (Rule 7.3). Press Save; the top of the Armory then says Saved, and Done closes it. You can change the choice until the Submit Deadline (Rule 7.4).



C.5 Submit. Press SUBMIT (4). The confirmation says the Bot goes in exactly as it is now; confirm it. SUBMIT waits while Pi is still working, and says so. The badge (5) shows Checking… while the check runs, then In followed by the Round's name when the Bot is accepted. If the badge shows Rejected, press the badge: the panel that opens shows what the check found. Ask Pi to fix it, practise, and submit again. You may submit as often as you like before the deadline; the newest accepted submission plays (Rule 11.5).

